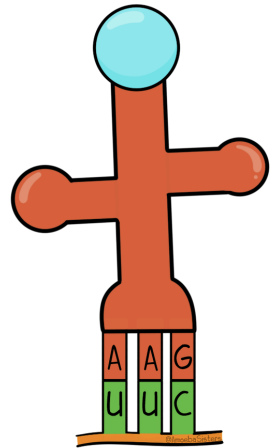


Amoeba Sisters Video Recap: How to Read a Codon Chart

Second Base

First Base	Second Base					Third Base	
	U	C	A	G			
	U	Phenylalanine	Serine	Tyrosine	Cysteine		U
		Phenylalanine	Serine	Tyrosine	Cysteine		C
		Leucine	Serine	STOP	STOP		A
		Leucine	Serine	STOP	Tryptophan		G
C	Leucine	Proline	Histidine	Arginine	U		
	Leucine	Proline	Histidine	Arginine	C		
	Leucine	Proline	Glutamine	Arginine	A		
	Leucine	Proline	Glutamine	Arginine	G		
A	Isoleucine	Threonine	Asparagine	Serine	U		
	Isoleucine	Threonine	Asparagine	Serine	C		
	Isoleucine	Threonine	Lysine	Arginine	A		
	Methionine	Threonine	Lysine	Arginine	G		
G	Valine	Alanine	Aspartic Acid	Glycine	U		
	Valine	Alanine	Aspartic Acid	Glycine	C		
	Valine	Alanine	Glutamic Acid	Glycine	A		
	Valine	Alanine	Glutamic Acid	Glycine	G		

1. Please *label* the following five words on the image below: **tRNA**, **mRNA**, **codon**, **amino acid**, and **anticodon**.



2. When reading a codon chart, *unless otherwise specified*, you generally use the three bases that are part of the:

CIRCLE ONE: tRNA anticodon mRNA codon DNA codon DNA anticodon

3. The codon chart demonstrates that more than one codon can potentially code for the same amino acid.

CIRCLE ONE: TRUE FALSE

4. The information in this rectangular codon chart could be represented in a different way (ex: circular codon chart).

CIRCLE ONE: TRUE FALSE

5. How many different codon *combinations* are shown on this codon chart? _____

6. How many different *types* of amino acids are shown on this codon chart? _____

Codon Example 1 from Video

mRNA Codon:
AUG

tRNA Anticodon:
UAC

Amino Acid:
methionine

Codon Example 2 from Video

Try working through this before checking your answers in the video!

mRNA Codon:
CCA

tRNA Anticodon:

7. _____

Amino Acid:

8. _____

Codon Example 3 from Video

Try working through this before checking your answers in the video!

mRNA Codon:
GUC

tRNA Anticodon:

9. _____

Amino Acid:

10. _____



Amoeba Sisters Video Recap: How to Read a Codon Chart

		Second Base					
		U	C	A	G		
First Base	U	Phenylalanine	Serine	Tyrosine	Cysteine	Third Base	U
		Phenylalanine	Serine	Tyrosine	Cysteine		C
		Leucine	Serine	STOP	STOP		A
		Leucine	Serine	STOP	Tryptophan		G
	C	Leucine	Proline	Histidine	Arginine		U
		Leucine	Proline	Histidine	Arginine		C
		Leucine	Proline	Glutamine	Arginine		A
		Leucine	Proline	Glutamine	Arginine		G
	A	Isoleucine	Threonine	Asparagine	Serine		U
		Isoleucine	Threonine	Asparagine	Serine		C
		Isoleucine	Threonine	Lysine	Arginine		A
		Methionine	Threonine	Lysine	Arginine		G
	G	Valine	Alanine	Aspartic Acid	Glycine		U
		Valine	Alanine	Aspartic Acid	Glycine		C
		Valine	Alanine	Glutamic Acid	Glycine		A
		Valine	Alanine	Glutamic Acid	Glycine		G

11. The video gave an example of identifying codons for an amino acid. Which six **mRNA codons** code for **leucine**?

12. What would be the six **tRNA anticodons** for the above mRNA codons?

13. What would be the **DNA** base triplets on the DNA template strand that complement the mRNA? One example has been filled in for you!

AAT

TRUE or FALSE?

14. _____ Changing one base letter on a **mRNA codon** will always change the amino acid it codes for.

15. Why or why not? _____

TRUE or FALSE?

18. _____ The mRNA codons GCU and GCC would share the **same tRNA anticodon**.

19. Why or why not? _____

TRUE or FALSE?

16. _____ The mRNA codon UGG is the only codon that codes for the amino acid tryptophan.

TRUE or FALSE?

17. _____ The mRNA codon UCU and AGU code for the same amino acid.

20. **Stop codons** can signal the end of a polypeptide. What are the three stop codons?

21. **AUG** is generally a **start codon**. What is the **amino acid** that AUG codes for?

